

- RAG Learning

2.2 Equations and Inequalities

OCR GCSE Mathematics (J560) — Paper 1, 2 & 3 — Foundation and Higher

2.2.1 Solving Linear Equations

Linear equation: An equation where the highest power of the unknown is 1. The goal is to find the value of the unknown that makes the equation true.

The golden rule: **whatever you do to one side, do the same to the other side.**

One-Step and Two-Step Equations

Worked Example: Solve $5x - 3 = 17$.

Add 3 to both sides: $5x = 20$

Divide both sides by 5: $x = 4$

Worked Example: Solve $\frac{2x + 1}{3} = 5$.

Multiply both sides by 3: $2x + 1 = 15$

Subtract 1: $2x = 14$

Divide by 2: $x = 7$

Equations with Unknowns on Both Sides

Collect all variable terms onto one side, then all constants onto the other:

Worked Example: Solve $5x + 2 = 3x + 10$.

Subtract $3x$ from both sides: $2x + 2 = 10$

Subtract 2: $2x = 8$

Divide by 2: $x = 4$

Worked Example: Solve $4(2x - 3) = 6x + 2$.

Expand: $8x - 12 = 6x + 2$

Subtract $6x$: $2x - 12 = 2$

Add 12: $2x = 14$

Divide by 2: $x = 7$

Worked Example: Solve $\frac{x + 3}{4} = \frac{2x - 1}{3}$.

Multiply through by 12 (LCM of 4 and 3): $3(x + 3) = 4(2x - 1)$

Expand: $3x + 9 = 8x - 4$

Collect terms: $9 + 4 = 8x - 3x$, so $13 = 5x$

$$x = \frac{13}{5} = 2.6$$

Exam Tip: Always check your answer by substituting back into the original equation. This takes 10 seconds and will catch arithmetic slips.

2.2.2 Forming and Solving Equations

Many real-world and geometry problems require you to write an equation before solving it.

Worked Example: The angles of a triangle are $3x$, $(2x + 10)^\circ$, and $(x - 4)^\circ$. Find x and the three angles.

Angles in a triangle sum to 180° : $3x + (2x + 10) + (x - 4) = 180$

Simplify: $6x + 6 = 180$

$6x = 174$, so $x = 29$

Angles: 87° , 68° , 25° . Check: $87 + 68 + 25 = 180$ ✓

Worked Example: A rectangle has length $(3x + 2)$ cm and width $(x - 1)$ cm. Its perimeter is 46 cm. Find its area.

Perimeter: $2(3x + 2) + 2(x - 1) = 46$

$6x + 4 + 2x - 2 = 46$

$8x + 2 = 46 \Rightarrow 8x = 44 \Rightarrow x = 5.5$

Length = $3(5.5) + 2 = 18.5$ cm, Width = 4.5 cm

Area = $18.5 \times 4.5 = 83.25$ cm²

Worked Example: A bag of sweets contains n sweets. Ali takes 5, then Ben takes half of what remains. 12 sweets are left. How many sweets were in the bag?

After Ali: $n - 5$

After Ben: $\frac{n - 5}{2} = 12$

$n - 5 = 24 \Rightarrow n = 29$

2.2.3 Linear Inequalities

An **inequality** shows that one expression is greater than (or less than) another. The solution is a range of values, not a single value.

Symbols: $<$ (less than), $\leq$ (less than or equal to), $>$ (greater than), $\geq$ (greater than or equal to).

Solving Linear Inequalities

Use the same inverse operations as equations, with one critical difference:

When you multiply or divide both sides by a negative number, you must reverse the inequality sign.

Worked Example: Solve $3x - 7 > 5$.

Add 7: $3x > 12$

Divide by 3: $x > 4$

Worked Example: Solve $-2x + 1 \leq 9$.

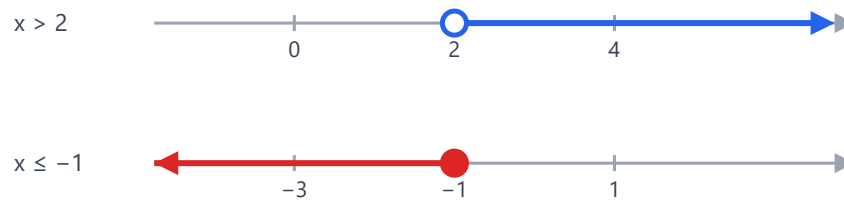
Subtract 1: $-2x \leq 8$

Divide by -2 (reverse sign!): $x \geq -4$

Representing Inequalities on a Number Line

Use an **open circle** $\circ$ for strict inequalities ($<$ or $>$) and a **filled circle** $\bullet$ for $\leq$ or $\geq$.

Number line representations



Combined Inequalities

E.g. $-3 < 2x + 1 \leq 7$: solve as two separate inequalities or work on all three parts simultaneously.

Worked Example: Solve $-3 < 2x + 1 \leq 7$ and represent on a number line.

Subtract 1 from all three parts: $-4 < 2x \leq 6$

Divide all three parts by 2: $-2 < x \leq 3$

Integer values: $x \in \{-1, 0, 1, 2, 3\}$

Number line: open circle at -2 , filled circle at 3 , line between them.

Exam Tip: When listing integer solutions, make sure to include the endpoints only if the sign is $\leq$ or $\geq$.

2.2.4 Simultaneous Equations

Two equations in two unknowns — find values of both unknowns that satisfy *both* equations simultaneously.

Method 1: Elimination

Add or subtract multiples of the equations to eliminate one variable.

Worked Example: Solve: $3x + 2y = 13$ and $5x - 2y = 11$.

Coefficients of y are $+2$ and -2 , so add the equations:

$$8x = 24 \Rightarrow x = 3$$

Substitute into equation 1: $9 + 2y = 13 \Rightarrow 2y = 4 \Rightarrow y = 2$

Check in equation 2: $15 - 4 = 11 \checkmark$

Worked Example: Solve: $2x + 3y = 16$ and $5x - y = 2$.

Multiply equation 2 by 3: $15x - 3y = 6$

Add to equation 1: $17x = 22 \Rightarrow x = \frac{22}{17}$

Substitute into equation 2: $y = 5x - 2 = \frac{110}{17} - \frac{34}{17} = \frac{76}{17}$

Method 2: Substitution

Rearrange one equation to express one variable in terms of the other, then substitute into the second equation.

Worked Example: Solve: $y = 2x - 1$ and $3x + 4y = 18$.

Substitute $y = 2x - 1$ into the second equation:

$$3x + 4(2x - 1) = 18$$

$$3x + 8x - 4 = 18 \Rightarrow 11x = 22 \Rightarrow x = 2$$

$$y = 2(2) - 1 = 3$$

Solution: $x = 2, y = 3$

Graphical Method

Plot both equations as straight lines. The solution is the point of intersection of the two lines.

- Parallel lines (same gradient, different intercept) $\rightarrow$ no solution.

- Coincident lines (same equation) → infinitely many solutions.

Exam Tip: In elimination, make sure the signs are correct when subtracting — this is the most common error. Write the subtraction out explicitly.

2.2.5 Quadratic Equations

A quadratic equation has the form $ax^2 + bx + c = 0$ and generally has **two solutions** (roots).

Method 1: Factorising

Rearrange to equal zero, factorise, then set each factor equal to zero:

Worked Example: Solve $x^2 - 5x + 6 = 0$.

Factorise: $(x - 2)(x - 3) = 0$

Either $x - 2 = 0$ or $x - 3 = 0$

$x = 2$ or $x = 3$

Worked Example: Solve $2x^2 + x - 6 = 0$.

Factorise (AC method: $ac = -12$; factors -3 and 4): $(2x - 3)(x + 2) = 0$

$x = \frac{3}{2}$ or $x = -2$

Method 2: The Quadratic Formula

For any quadratic $ax^2 + bx + c = 0$:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The expression $b^2 - 4ac$ is called the **discriminant**:

Value of $b^2 - 4ac$	Number of real roots	Nature of roots
$b^2 - 4ac > 0$	2 distinct real roots	Two different solutions
$b^2 - 4ac = 0$	1 repeated root	One solution (parabola touches x -axis)
$b^2 - 4ac < 0$	No real roots	Parabola does not cross x -axis

Worked Example: Solve $3x^2 - 5x - 2 = 0$ using the quadratic formula.

$$a = 3, b = -5, c = -2$$

$$b^2 - 4ac = 25 - 4(3)(-2) = 25 + 24 = 49$$

$$x = \frac{5 \pm \sqrt{49}}{6} = \frac{5 \pm 7}{6}$$

$$x = \frac{12}{6} = 2 \text{ or } x = \frac{-2}{6} = -\frac{1}{3}$$

Worked Example: Solve $x^2 + 3x - 1 = 0$, giving answers to 2 decimal places.

$$a = 1, b = 3, c = -1$$

$$b^2 - 4ac = 9 + 4 = 13$$

$$x = \frac{-3 \pm \sqrt{13}}{2}$$

$$x = \frac{-3 + 3.6056}{2} = 0.30 \text{ (2 d.p.) or } x = \frac{-3 - 3.6056}{2} = -3.30 \text{ (2 d.p.)}$$

Exam Tip: The quadratic formula always works, even when factorising seems impossible. Always rearrange to $ax^2 + bx + c = 0$ first. Identify a, b, c before substituting — include the sign.

Higher — Completing the Square

Write $ax^2 + bx + c$ in the form $a(x + p)^2 + q$. This reveals the vertex of the parabola and is a third method for solving quadratics.

For $x^2 + bx + c$ (coefficient of x^2 is 1):

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2 + c$$

Worked Example: Solve $x^2 + 6x + 2 = 0$ by completing the square.

$$(x + 3)^2 - 9 + 2 = 0$$

$$(x + 3)^2 = 7$$

$$x + 3 = \pm\sqrt{7}$$

$$x = -3 \pm \sqrt{7}$$

Worked Example: Write $2x^2 - 8x + 5$ in the form $a(x + p)^2 + q$. Hence state the minimum value and the x -coordinate where it occurs.

$$\text{Factor out 2: } 2(x^2 - 4x) + 5$$

$$\text{Complete the square inside: } 2[(x - 2)^2 - 4] + 5$$

$$\text{Expand: } 2(x - 2)^2 - 8 + 5 = 2(x - 2)^2 - 3$$

$$\text{Minimum value is } -3 \text{ at } x = 2$$

2.2.6 Algebraic Proof [Higher]

Higher only

Algebraic proof uses algebra to show that a statement is always true. You must work with general expressions, not specific numbers.

Key vocabulary:

- Even number: $2n$
- Odd number: $2n + 1$
- Consecutive integers: $n, n + 1, n + 2, \dots$
- Multiple of k : kn (where n is any integer)

Showing Identities

Worked Example: Prove that $(n + 3)^2 - (n + 1)^2 \equiv 4(n + 2)$.

$$\begin{aligned}\text{LHS: } & (n + 3)^2 - (n + 1)^2 \\ &= n^2 + 6n + 9 - (n^2 + 2n + 1) \\ &= n^2 + 6n + 9 - n^2 - 2n - 1 \\ &= 4n + 8 = 4(n + 2) = \text{RHS } \checkmark\end{aligned}$$

Proving Divisibility

Worked Example: Prove that the sum of any three consecutive integers is always divisible by 3.

$$\begin{aligned}\text{Let the three integers be } & n, n + 1, n + 2 \\ \text{Sum} = & n + (n + 1) + (n + 2) = 3n + 3 = 3(n + 1) \\ \text{This is a multiple of 3, so the sum is always divisible by 3. } & \square\end{aligned}$$

Worked Example: Prove that the product of two odd numbers is always odd.

$$\text{Let the two odd numbers be } 2m + 1 \text{ and } 2n + 1$$

$$\text{Product} = (2m + 1)(2n + 1) = 4mn + 2m + 2n + 1 = 2(2mn + m + n) + 1$$

This is in the form $2k + 1$ (where $k = 2mn + m + n$ is an integer), so the product is odd. $\square$

Disproving by Counterexample

To disprove a general claim, you only need to find ONE example where it fails — a **counterexample**.

Worked Example: Disprove: "The sum of two prime numbers is always even."

Counterexample: $2 + 3 = 5$ (odd). $\square$

Exam Tip: In a proof question, use fully general algebraic expressions. Do not use specific numbers. End with a clear concluding sentence explaining what you have shown.

Quick Reference Summary

Topic	Key method / formula
Linear equations	Inverse operations; same operation on both sides
Inequalities	Same as equations but reverse sign when multiplying/dividing by a negative
Simultaneous (linear)	Elimination (match coefficients) or substitution
Quadratic: factorising	$(x - p)(x - q) = 0 \Rightarrow x = p \text{ or } x = q$

Quadratic formula	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
Completing the square [H]	$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2 + c$
Algebraic proof [H]	Use $2n$ (even), $2n + 1$ (odd), $n, n + 1, \dots$ (consecutive)